

Clinical Features

In most patients, the onset of alopecia areata occurs within the first three decades of life, although it can begin at any age. The incidence is approximately equal between sexes.

Etiology and Pathogenesis

Alopecia areata is a chronic, organ-specific autoimmune disorder, primarily mediated by autoreactive T cells, targeting hair follicles and occasionally the nails. There is an increased prevalence of other autoimmune conditions—particularly autoimmune thyroid disease—in affected adults, as well as a higher frequency of organ-specific autoantibodies. While hair follicle autoantibodies have been detected in some patients, they are unlikely to play a primary role in disease pathogenesis.

Evidence from studies using human hair follicles transplanted into immunodeficient mice supports a cell-mediated mechanism, with **CD8⁺ T cells** playing a central role. These cells may recognize autoantigens presented by major histocompatibility complex class I molecules, possibly expressed during anagen or anagen-associated melanogenesis. This process is thought to be triggered by a collapse of the normal immune privilege of the anagen hair bulb, potentially induced by interferon- γ . This loss of immune privilege may expose sequestered melanogenesis-related autoantigens, which then become targets of autoimmune attack—an idea supported by accumulating experimental and clinical evidence.

Histologically, alopecia areata is characterized by a peribulbar inflammatory infiltrate composed predominantly of T cells around anagen hair follicles. However, histopathologic findings are stage-dependent, and classic features may be absent in subacute or chronic lesions. Therefore, alopecia areata should

remain in the differential diagnosis even when the typical lymphocytic cuff is missing, especially if there is a high proportion of telogen or miniaturized hairs.

Anagen follicles under immune attack prematurely enter catagen and then telogen, eventually reaching a dystrophic anagen state. Although the follicles continue to cycle, active disease prevents progression beyond early anagen, resulting in failure to produce normal hair shafts. Importantly, hair follicles are not destroyed in alopecia areata, likely because the follicular stem cell compartment in the mid-portion of the outer root sheath is spared.

Genetic predisposition plays a significant role in alopecia areata. Approximately 10–20% of patients report a family history of the condition, with the actual heritability likely higher due to underreporting of mild or cosmetically concealed cases. Cases of concordant alopecia areata in twins, including those with simultaneous onset, further support a genetic component.